

Article

Assisting Hearing and Physically Impaired Students in Navigating Immersive Virtual Reality for Library Orientation

Pakinee Ariya ¹, Yakannut Yensathit ², Phimpakan Thongthip ², Kannikar Intawong ³ and Kitti Puritat ^{2,*} 

¹ College of Art and Technology, Chiang Mai University, Chiang Mai 50200, Thailand; pakinee.a@cmu.ac.th

² Department of Library and Information Science, Faculty of Humanities, Chiang Mai University, Chiang Mai 50200, Thailand; yakannut_y@cmu.ac.th (Y.Y.); phimpakan_t@cmu.ac.th (P.T.)

³ Faculty of Public Health, Chiang Mai University, Chiang Mai 50200, Thailand; kannikar.i@cmu.ac.th

* Correspondence: kitti.p@cmu.ac.th

Abstract: This study aims to design and develop a virtual reality platform (VR-ISLS) tailored to support hearing and physically impaired students at the university library for navigating and utilizing library services. By employing an immersive virtual environment, the platform replicates the physical setting of the university's library to create a realistic experience that reduces anxiety and enhances familiarity. The platform integrates assistive technology functions, including sign language interpretation, customizable audio cues, vibration feedback, and various locomotion controls to meet the diverse needs of impaired students. The research methodology employs an iterative development process, incorporating feedback from library staff, disability support services, and students to ensure usability and accessibility. Evaluation of the platform using the System Usability Scale (SUS) and user feedback revealed a positive reception, with recommendations for further customization and enhanced assistive features to optimize the user experience. This study underscores the importance of inclusive design and continuous iteration in creating immersive virtual reality tools that provide significant benefits for persons with disabilities, enhancing both accessibility and learning experiences.



Academic Editor: Reza Saatchi

Received: 21 November 2024

Revised: 19 December 2024

Accepted: 21 December 2024

Published: 24 December 2024

Citation: Ariya, P.; Yensathit, Y.; Thongthip, P.; Intawong, K.; Puritat, K. Assisting Hearing and Physically Impaired Students in Navigating Immersive Virtual Reality for Library Orientation. *Technologies* **2025**, *13*, 2. <https://doi.org/10.3390/technologies13010002>

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: virtual reality; assistive technology; hearing impairment; physical impairment; accessibility; immersive environment

1. Introduction

In recent years, virtual reality has grown in popularity as a novel technology [1]. It works by generating a simulated environment in which users can engage in immersive experiences that mimic real-world interactions [2]. According to Schutte and Stilinović [3], virtual reality's capacity to provide a sense of presence and gaining perspective enhances users' connection with others and understanding of diverse perspectives. Consequently, researchers discovered that virtual reality effectively increases empathy, allowing people to better relate to and understand the experiences, emotions, and motivations of others [4]. Disability-related fields have employed virtual reality, primarily for medical treatment and rehabilitation, rather than advocacy. For instance, it has been utilized for teaching people with autism emotion perception and social cognition [5]. It has also been employed for helping individuals who have intellectual disabilities in developing daily living skills [6], as well as for physical fitness training, sports, motor skill development, and recovery of muscles for individuals with developmental and physical disabilities [7]. One important component of adopting virtual reality for disability advocacy is that it emphasizes users' comprehension and interaction with their environment. Rather than imitating an impaired

body, virtual reality emphasizes and clarifies the barriers presented by inaccessible locations, communities, or the world. While these applications are significant, they only address the medical and rehabilitation aspects of virtual reality. The social applications of virtual reality, notably for campaigning on behalf of individuals with disabilities, are mostly undiscovered. Despite its intriguing potential for this objective, particularly from a disability studies viewpoint, virtual reality remains unexplored in the field of disability advocacy [8]. Further study is needed to fully grasp virtual reality's potential as an advocacy tool for disabled people. Therefore, the purpose of this study is to develop and enhance virtual reality, particularly for people with hearing impairments, with an emphasis on assisting them with everyday tasks.

This paper addresses these gaps by first reviewing the literature on virtual reality technologies for individuals with disabilities (Section 2), highlighting research gaps and design opportunities. Section 3 outlines the study's objectives and the development of the VR-ISLS platform for hearing and physically impaired students. Section 4 describes the iterative design process, followed by Section 5 detailing the methodology, including participants, ethical considerations, and data collection. The findings, including usability evaluations and user feedback, are presented in Section 5.3, while Section 6 discusses these results in the context of existing research. Finally, Section 7 concludes with the study's contributions, limitations, and recommendations for future research and development, providing a comprehensive exploration of VR-ISLS for accessibility in academic libraries.

2. Related Works

2.1. Virtual Reality (VR)

Virtual reality serves as a computer-generated simulation of reality. Utilizing a video headset, computer programs generate a visual environment that can replicate the real world, down to the pixel—or present an entirely fantastical realm [9]. Various researchers have offered definitions for virtual reality. One such definition characterizes virtual reality as a computer-generated representation of reality. In essence, computer programs display a visual environment through a video headset, capable of accurately simulating the real world or presenting an entirely artificial one [9]. Virtual reality, according to Lavalle, a professor in the Department of Computer Science at the University of Illinois at Urbana-Champaign, is the use of artificial sensory stimulation to elicit specific behaviors in an organism without the organism being aware of the external influence [10]. Similar to other forms of media, virtual reality possesses the capacity to educate, captivate, enlighten, influence, and persuade. Of specific relevance is its role in facilitating learning, especially in youth-oriented library programs, as well as in broader initiatives targeting adults to stimulate interest and encourage repeated patronage for accessing information through the library system. Libraries have been at the forefront of the makerspace movement, establishing public spaces conducive to informal learning where users, spanning a diverse range, can engage with new technologies [11]. Digital media, technology, and maker-space initiatives represent crucial elements in the institutional evolution of libraries, contributing to their success as venues for youth involvement. Additionally, these endeavors play a significant role in enhancing technology access and equity for adults within library settings [12]. Educational achievements in virtual reality (VR) have been linked to the degree of a VR user's sense of presence during an encounter. A VR environment that fosters a robust sense of presence holds the potential to impact the learning process and the sharing of information related to locations, individuals, objects, and experiences by immersing users in simulated "lived experiences [13]". Within VR experiences, content creators create elaborate virtual environments and develop narratives that are both immersive and interactive. These narrative-driven and frequently interactive stories are recognized

as potent means of leveraging media for educational purposes and altering perspectives on challenging topics, such as attitudes toward individuals facing homelessness or racial stereotypes [4].

2.2. Hearing Impaired

The fundamental concept of disability studies is that disability should not be considered as a physical or mental limitation. Instead, it should be viewed as a reflection of society's failure to accommodate people with disabilities, resulting in considerable disadvantages caused by the surrounding environment and attitudes [14]. Disability is more than solely a health concern; it is a complicated phenomenon caused by the connection of an individual's physical or mental aspects with the social setting in which they live. To address the issues that persons with disabilities encounter, society must strive to remove environmental and societal obstacles. A person's inability to make decisions about meeting their fundamental requirements or fully engaging in social life is not exclusively due to their medical or mental condition, whether present at birth or developed later [15]. Disabling hearing loss can be described as a loss that exceeds 40 decibels in adults and 30 decibels in children. The decibel (dB) is a unit used to quantify the loudness of sound or an electrical signal's power level through a comparison to a logarithmic reference level. The World Health Organization (WHO) classifies hearing impairment into four categories: mild (26 to 40 dB), moderate (41 to 60 dB), severe (61 to 80 dB), and profound (above 81 dB), based on pure tone average levels [16]. Hearing impairment can result from environmental or genetic factors, though in many cases, the exact cause remains undetermined [17]. Hearing loss ranks as the fourth leading cause of disability worldwide, and its prevalence is on the rise. The World Health Organization (WHO) predicts that by 2050, over 900 million individuals around the globe will experience some degree of hearing loss [16]. In low- and middle-income countries, environmental factors such as meningitis, measles, and ototoxicity are the primary causes of hearing impairment, whereas their impact is less significant in high-income countries. This disparity is largely due to inadequate healthcare systems in lower-income regions, which often lack the resources to effectively prevent, screen, and manage the hearing impairment causes [18].

Students with hearing impairments experience difficulties that can hinder their learning processes. Those who are deaf from an early age tend to have lower educational performance than those who lose their hearing later in life. For example, their vocabulary and reading skills may be significantly affected [15]. Students with hearing loss tend to favor a visual learning method. They may struggle to learn effectively when information is only delivered verbally. FM systems can be useful tools for them while in classes [19]. The effects of being hard of hearing are often evident in terms of time. Students who need recorded information typically experience delays in receiving it, which can lead to various issues, including a decline in their self-esteem and confidence.

2.3. Physical Disabilities

Physical disabilities are conditions that restrict a person's ability to engage in daily activities [20]. Physical disability refers to any impairment or limitation that restricts the function of one or more limbs. This condition can be either temporary or permanent and can result from various causes, such as accidents, injuries, illnesses, post-surgical effects, or hereditary factors [21]. Physical disabilities can stem from acquired, congenital, or genetic factors. An individual with a congenital or hereditary physical disability has experienced disability from birth, either because of an injury sustained during childbirth or because of inherited genetic disorders or difficulties with muscle cells. Moreover, physical disability may arise for a variety of causes. Severe accidents, brain damage, infections, illnesses, and

the aftermath of disorders and other medical problems, including dementia and stroke, can all cause them [22]. Table 1 provides examples of physical disabilities.

Table 1. The examples of physical disabilities.

Physical Disabilities	Description
Limb loss and amputations	A part of the body has been removed in part or whole. Amputations can result from surgery, animal attacks, accidents, and war [23].
Arthritis	Arthritis and musculoskeletal disorders can affect muscles, bones, and joints. Common conditions include osteoarthritis, back pain, rheumatoid arthritis, fibromyalgia, osteoporosis, gout, lupus, and ankylosing spondylitis. It affects people of all ages, ethnicities, and lifestyles, including children and teenagers [23].
Birth defects	An abnormality that can be found during pregnancy, at birth, or in early life is called a birth defect. The abnormality can alter the baby's metabolism or have an impact on the structure or function of a bodily component. Certain congenital abnormalities, including the absence or malformation of a certain bodily component, may need continuous medical treatment or therapy [23].
Cerebral palsy	Several disabilities related to posture and movement are all referred to as cerebral palsy. When they develop and practice motor skills, a kid with cerebral palsy may eventually gain more control over their movements. The illness is not passed on or inherited. It is brought on by a brain injury [23].
Cystic fibrosis (CF)	A potentially fatal genetic condition is cystic fibrosis (CF). The digestive system and lungs are badly impacted. A kid with cystic fibrosis inherits two copies of a defective gene, one from each parent. Most cases of CF are identified at birth. It affects both men and women equally and is not communicable [23].

2.4. Virtual Reality for Disabilities

Fifteen percent of the world's population comprises individuals living with disabilities. These individuals often encounter challenges in their daily lives, including communication difficulties, limited access to healthcare services, transportation issues, and obstacles to community mobility. In other words, individuals with disabilities are often viewed as a marginalized community. Limitations on their access to community support services and healthcare can significantly worsen their daily challenges [24]. Several researchers have employed virtual reality (VR) to improve access to public services and assist people with disabilities in overcoming obstacles in daily life as described in Table 2.

Table 2. The overview of studies on VR to support people with disabilities.

Year	Project Name	Objective	Technology	Summary
2022	Learn to Sail	To present a short summary of the findings of a study that examined the sailing knowledge and abilities of people with physical disabilities	virtual reality sailing simulation	Seven participants with disabilities indicated that the training helped them feel more capable of learning how to sail and that it fulfilled their desire to sail on water [25].
2022	N/A	To design a VR technology to evaluate pitch ranking capabilities among kids with cochlear implants, hearing aids, or ordinary hearing	VR	VR appeared beneficial in measuring pitch ranking in kids despite their hearing condition and had great promise to evaluate or teach auditory capabilities such as localization or perception of music tasks [26].
2022	N/A	To design a VR technology to evaluate pitch ranking capabilities among kids with cochlear implants, hearing aids, or ordinary hearing	VR	VR appeared beneficial in measuring pitch ranking in kids despite their hearing condition and had great promise to evaluate or teach auditory capabilities such as localization or perception of music tasks [26].
2023	N/A	To increase the potential of group-based virtual reality gaming in reducing loneliness and social isolation among teenagers with physical disabilities	VR games	It can improve the mental health of teenagers with disabilities. Using affordable, accessible consumer electronics, participants formed friendships with peers and other online players [27].

However, research has rarely addressed the specific needs of hearing and physically impaired students in educational environments, such as university libraries. This discussion highlights the gaps in accessibility that this study seeks to address, as outlined in the research questions presented in the following section.

3. Purpose of the Study

The primary purpose of this study is to design and develop a platform to support hearing and physically impaired students at Chiang Mai University. This research focuses on creating assistive technology and functions in virtual reality (VR) specifically tailored to these students to facilitate their understanding of library services and enhance their academic experience at the university, as provided by the Chiang Mai University Library (CMUL). To address the challenges that hearing, and physically impaired students encounter when accessing library resources, this study aims to develop VR-ISLS (virtual reality for impaired students in library services), tailored to their unique needs. The research addresses the following questions:

RQ1: How can immersive virtual reality be designed to support orientation for the main university library and assist hearing and physically impaired students?

RQ2: How do hearing and physically impaired students perceive the usability and accessibility of the VR-ISLS platform?

RQ3: What are the essential design features of a virtual reality platform tailored to hearing and physically impaired students to effectively simulate library orientation?

These research questions guide the development of an immersive VR system that addresses both the practical and accessibility needs of hearing and physically impaired students. By exploring these questions, the study will identify effective methods for designing a VR platform that supports library orientation, ensuring that hearing and physically impaired students can navigate and access services independently. Investigating user perceptions of usability and accessibility will provide direct feedback from end-users, enabling refinements to meet their specific needs. Pinpointing essential design features will lead to a functional, inclusive tool that enhances independence, supports academic success, and significantly improves the educational experience of hearing and physically impaired students at the university.

4. The Design Process of Virtual Reality for Impaired Students

Since the COVID-19 pandemic in 2020 [28], digital technology has been increasingly used to promote library services in academic libraries, replacing traditional library orientations for first-year students with augmented reality [29], virtual reality [30], video clips [31], and Zoom meetings [32]. However, most of these digital approaches have not been designed with the physical needs of impaired students in mind. Therefore, the purpose of this research project is to investigate the feasibility of using virtual reality as an alternative method to promote library services to all impaired students at Chiang Mai University.

The development of VR-ISLS (virtual reality for impaired students in library services) was initiated by assessing how VR can help overcome physical and sensory barriers in traditional library settings, aiming to make library resources more accessible and less intimidating for these students. This collaborative effort involves the Chiang Mai University Library (CMUL), which provides the visual environment and library services; the Department of Library and Information Science (LIS), which offers expertise in library anxiety and the principles of library services; the Disability Support Services (DSS), which supports impaired students at Chiang Mai University; and the Game in Information System and Digital Cultural Heritage Lab (GDCH), which develops the VR platform based on the digital twin concept of the Chiang Mai University Library. This multidisciplinary approach aims to create a supportive and innovative environment where impaired students can access and use library services without stress, enhancing their academic experience and learning outcomes. The design process was adapted from the virtual reality development method [33], which consists of four phases: concept, preparation and design, implementa-

tion, and evaluation. Each of these phases is described in detail in the following sections, with the overall process illustrated in Figure 1.

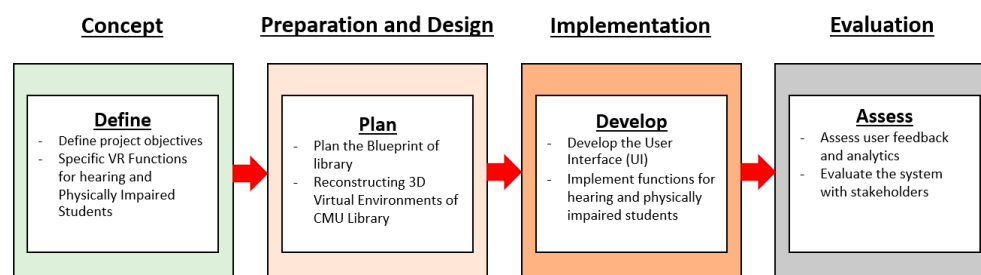


Figure 1. The overall process of implementing the VR application.

4.1. Concept Phase

In the concept phase, the primary goal is to define the project's objectives and develop a clear understanding of the needs of impaired students at Chiang Mai University who will be using the virtual reality for impaired students in library services. This study focuses on students with hearing and physical impairments. This phase begins with a detailed information-gathering process, during which the development team engages with key stakeholders, including librarians from the Chiang Mai University Library, experts from the Department of Library and Information Science, personnel from Disability Support Services, and developers from the Game in Information System and Digital Cultural Heritage Lab. Through interviews, surveys, and observation sessions, the team gains insights into how these students interact with library services and identifies the barriers they face. The findings, which summarize the user requirements, including the types of impairments present at Chiang Mai University and specific user concerns, are presented in Table 3.

The team then outlines the project scope, functional requirements, and the virtual environment. Regarding the virtual environment, VR-ISLS incorporates the concept of a digital twin in academic libraries [34,35]. A digital twin is a virtual representation of a real, physical object or environment. In this case, the digital twin serves as a digital representation of the planned or existing physical library at the Chiang Mai University Library, which serves students from all faculties at Chiang Mai University. It replicates the layout, design, and functionality of the physical library, allowing users to interact with it in a virtual environment. Using the digital twin concept, users can navigate and explore the virtual library, simulating the experience of being physically present in the library. In terms of functional requirements, and considering the cost budget, time limitations, and project complexity, we plan to implement support for only two main types of impairments, excluding visual impairment.

Table 3. Types of impairments of students in university, user concerns, and specific VR functions.

Main of Impairment	Type of Characteristic	User Concerns	Specific Functions in VR
Hearing Impairment	Sensorineural Hearing Loss [36,37]	Difficulty hearing sounds clearly; trouble understanding speech in noisy environments; and often permanent	Enhanced audio settings, visual cues and subtitles, and vibrational feedback
	Conductive Hearing Loss [37,38]	Blockage or malformation in outer/middle ear affecting hearing; and temporary or fluctuating hearing loss	Adjustable volume, visual cues and subtitles, and vibrational feedback

Table 3. *Cont.*

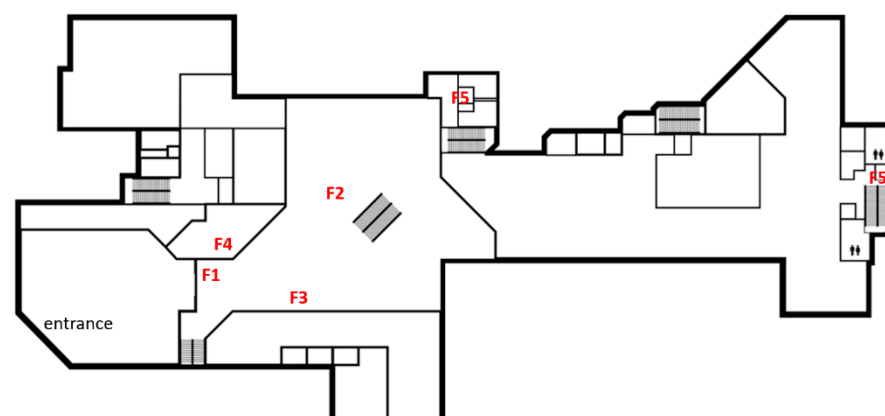
Main of Impairment	Type of Characteristic	User Concerns	Specific Functions in VR
Physical Impairment	Mixed Hearing Loss [37,39]	Combination of conductive and sensorineural issues; varied severity and symptoms; and challenges with both sound clarity and volume	Sign language support, visual cues and subtitles, and vibrational feedback
	Profound Hearing Loss (Deafness) [37,40]	Inability to hear any sounds; reliance on visual and tactile communication; and complete dependence on non-auditory cues	Sign language support, visual cues and subtitles, and vibrational feedback
	Mobility Impairments [41]	Difficulty with movement and control of VR interfaces; need for adaptive controls; and fatigue from prolonged use of traditional controllers	Alternative locomotion controller system, adjustable VR interaction zones, and adaptive controllers
	Upper or Lower Limb Amputation [42]	Limited or no use of one or more limbs; challenges with traditional input devices; and need for customization based on the level of amputation	Alternative locomotion controller system, adjustable VR interaction zones, and adaptive controllers for one hand only
	Muscular Dystrophy [43]	Progressive muscle weakness and loss of muscle mass; difficulty with precise movements and stamina; and fatigue and need for frequent breaks	Alternative locomotion controller system and adjustable VR interaction zones

4.2. Preparation and Design Phase

The preparation and design phase focuses on establishing the necessary virtual environment and collecting data to prepare the information and tools needed to develop the VR application. During this phase, the team works on designing the 3D environment based on the blueprint of Chiang Mai University Library and identifying key points to include in the VR environment. They collect data on specific library services for impaired students to ensure that these services provide the necessary knowledge, as summarized in Table 4. This process involves creating visual materials and 3D environments to refine the visual and interactive aspects of the VR scenes, thereby enhancing their accessibility and the accuracy of information for students with impairments.

Table 4. Physical environment of the library for hearing and physically impaired students.

Physical Environment for Impaired Students	Reference Point (Figure 2)	Description
Accessible Entrance	F1	The entrance door is wide enough to accommodate wheelchairs, and staff are available at the entrance at all times.
OPAC Search Computers	F2	The desk height is suitable for wheelchairs, and the chairs have wheels, making them easy to move.
Seating Area for Impaired Students	F3	This area has adequate lighting, is easily accessible as it is near the entrance, and has desks at wheelchair height with movable chairs.
Information and Loan/Return Counters	F4	Staff are specifically available to assist impaired students with tasks such as finding books or answering academic inquiries.
Accessible Restroom	F5	The restroom is equipped with wide doors, handrails, and emergency call buttons. The space inside is sufficient for maneuvering wheelchairs comfortably.

**Figure 2.** The blueprint of the first floor and physical environment of the library for hearing and physically impaired students.

Simultaneously, there is a collaborative effort with the library staff to develop engaging content for the VR experience. This includes creating digital versions of library guides, instructional videos with sign language interpreters, and interactive quizzes designed to make the library's resources more accessible and to enhance the learning experience for impaired students. In addition to content development, the team selects the most suitable VR hardware, such as the Meta Quest 2, and software platforms such as the Unity game engine version 2022.1.8, which is compatible with the Meta XR SDK. This selection is made to develop a 3D virtual reality environment that meets the specific needs of impaired students and fulfills the overall project requirements.

Finally, the team designed the 3D environment to simulate the actual Chiang Mai University Library based on the concept of a digital twin, which includes 3D modeling and scanning by the photogrammetry method [44,45]. The environment is designed to include various objects in the library, such as architectural elements, natural objects, and specific

areas such as the front room and the space room. This approach provides a realistic and immersive representation of the physical environment, aimed at reducing user anxiety. The blueprints of the first floor of the library, including reference points within the library where users can participate in learning, are shown in Figures 2 and 3. Additionally, the library services information provided by Chiang Mai University Library is shown in Table 5.



Figure 3. The 3D environment of the library and points of interest for user participation.

Table 5. Services provided for students with impairments at Chiang Mai University.

Library Services	Description
Extended Loan Period for Information Resources	General Thai and foreign language books can be borrowed for 30 days, theses for 14 days, novels for 14 days, audio-visual materials for 14 days, and CDs/DVDs for 14 days.
Book Retrieval Service	This service is specifically for students with impairments to ensure they can access and use information resources and knowledge in the library on an equal basis with other students. The service includes retrieving books from various shelves by library staff upon request at the loan-return counter.
Library Orientation Service	This service is offered within the library to introduce students to all areas across the library's four floors and to various information resources and basic services available within the library.
Document Delivery Service	This service delivers books to students with impairments of all types who need access to information and library resources for borrowing. Students can request books from faculty libraries to be delivered to the central library or the desired faculty library through an online service.
Book Delivery Service to University Dormitories	This service delivers books to students who need library resources from the central library and are unable to travel to the library. Books can be delivered to dormitories within Chiang Mai University by staff, available Monday to Friday from 8:30 a.m. to 4:00 p.m. Details of the books and the appointment for receiving the books must be arranged with the staff through an online channel.

4.3. Implementation Phase

This phase focuses on programming the core functionality of the VR application, integrating the content developed in earlier phases, and implementing navigational aids and accessibility features to ensure a fully accessible environment for students with impairments. For functions related to hearing impairments, special attention is given to integrating user interfaces for instructional videos with sign language interpreters and subtitles in both Thai and English, as well as an audio cue function that allows customization of volume levels, as shown in Figure 4. The team ensures that these interfaces are seamlessly integrated into the VR application, providing users with an intuitive and cohesive experience. Additionally, vibrational feedback is delivered through the controller when students approach an interaction point, and an adjustable volume function is included for students with sensorineural hearing loss. This integration is essential for maintaining the system's usability and accessibility, particularly for students with hearing impairments.

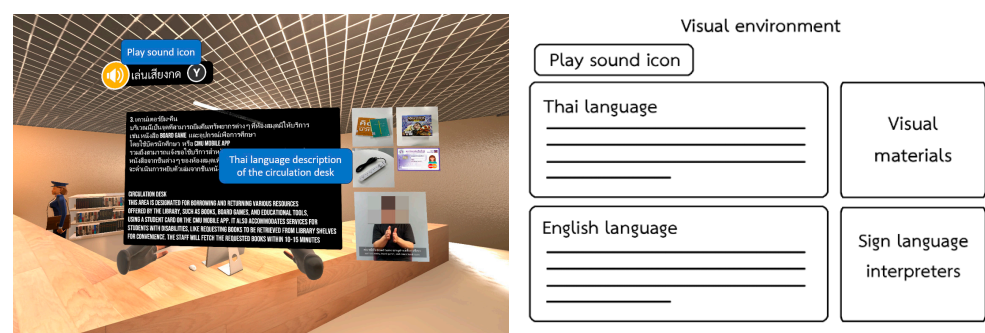


Figure 4. Examples of user interfaces (left) and the design of a user interface for impaired students (right).

In terms of functionality for physical impairment, we have developed three types of alternative locomotion controller systems: locomotion with the gamepad (controller-based technique) [46], locomotion with teleportation (teleportation-based technique) [47], and locomotion with a hybrid method (combining gamepad and head tracking). The overview of these locomotion techniques is shown in Figures 5 and 6.



Figure 5. Example of locomotion with the gamepad (left) and locomotion with teleportation (right).

The first system, locomotion with the gamepad, uses movement control with a thumbstick on the touch controller. The thumbstick on the left controller is used for translational movement: forward, backward, left, and right. The thumbstick on the right controller is used for turning to achieve different angles of view. The second system, locomotion with teleportation, involves discontinuous movement by selecting a position on the ground and then moving to that position. The user needs to press and hold the right trigger button to specify the teleportation anchor, determining where to move. A direction controller then appears and can be aimed in a direction using the right thumbstick.

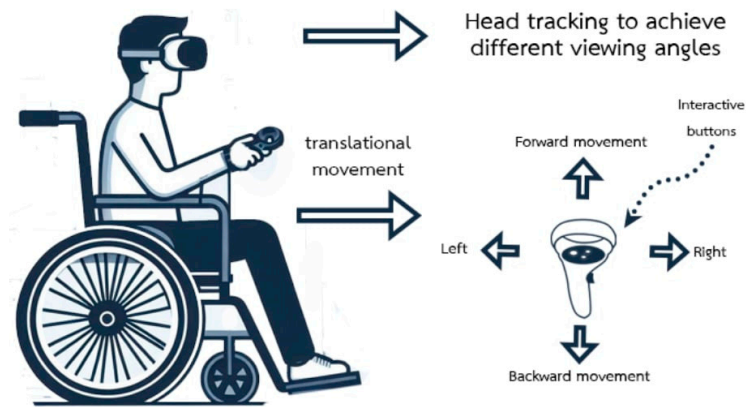


Figure 6. Example of a hybrid method (combining gamepad and head tracking) for students with upper limb amputations.

The third system, locomotion with a hybrid method, is designed for impaired students who can use only one hand. This system uses movement control with a thumbstick on the touch controller. The thumbstick on either the right or left controller is used for translational movement: forward, backward, left, and right. Turning is achieved through head tracking, utilizing the gyroscope sensor in the virtual reality headset. It is important to note that users can switch to using either the right- or left-hand controller for translational movement.

4.4. Evaluation Phase

The evaluation phase is critical in assessing the VR-ISLS application for bugs, usability, knowledge provision, and physical environment. Extensive user testing is conducted to gather feedback from librarians, lecturers, experts from Disability Support Services, and impaired students who utilize the application. The process begins with pilot testing to identify major issues, followed by usability testing where students interact with the VR environment while observers note ease of use, navigation, and challenges faced. A thorough accessibility assessment ensures the application meets the needs of all impaired students by verifying compliance with established accessibility guidelines and conducting functionality tests for features such as sign language interpreters, alternative locomotion controls, and customizable sensory settings. This comprehensive evaluation helps ensure that the VR-ISLS application is effective, user-friendly, and accessible to all students.

5. Research Methodology

This study employs a research and development (R&D) methodology to design and evaluate the virtual reality for VR-ISLS platform at Chiang Mai University. The research focuses on developing assistive technologies and functions specifically for students with hearing and physical impairments to help them navigate library resources. Given the inclusion of a vulnerable population, this project was reviewed and approved by the Chiang Mai University Research Ethics Committee (CMU REC No. 66/169). The project was conducted in collaboration with key stakeholders, including library staff, the Office of Disability Support Services, and faculty members from the Department of Library and Information Science, ensuring comprehensive support and accessibility considerations throughout the development.

5.1. Participants

According to data from Disability Support Services, Chiang Mai University has fifty students with disabilities spanning three primary types of impairments across all 11 faculties. While all impaired students were invited to participate, those with visual impairments were excluded due to the scope and limitations of the application's functionality. In addi-

tion, exclusion criteria included severe motion sensitivity or cybersickness, neurological conditions (e.g., epilepsy), significant cardiovascular issues, and discomfort with VR technology. Ultimately, 20 students agreed to participate. The demographic and characteristic details of these students are provided in Table 6. Notably, in terms of library usage frequency, 35% (N = 7) of participants indicated that they “rarely” visit the library, while 65% (N = 13) reported they “never” visit.

Table 6. Demographics of participants (N = 20).

Impairments	Impairment Characteristics	Symptom Details	Gender
Hearing Impairment (N = 8)	Complete hearing loss	Individuals with profound deafness who communicate primarily through sign language	Male (N = 1), Female (N = 1)
	Hearing Aid Tool	Individuals with partial hearing impairment who often utilize hearing aids or similar devices for auditory support	Male (N = 2), Female (N = 4)
Physical Impairment (N = 12)	Physical constraint using wheelchair	Individuals who rely on a wheelchair to assist with mobility	Male (N = 0), Female (N = 2)
	Physical constraint with muscle weakness but without using a wheelchair	People experiencing severe muscle weakness that impacts both mobility and physical tasks	Male (N = 2), Female (N = 6)
	Upper limb amputation	Individuals who have functional use of one hand	Male (N = 1), Female (N = 1)

5.2. Instrument for Measurement

5.2.1. System Usability Scale

The System Usability Scale (SUS) [48] was used as a standardized tool to evaluate the overall usability of the virtual reality system [49]. The SUS is a widely recognized 10-item questionnaire designed to capture user perceptions of system usability on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). It measures aspects such as ease of use, integration of features, and the user’s confidence in navigating the virtual environment. Given the accessibility-focused design of VR-ISLS, the SUS was slightly modified to reflect the assistive functions specific to hearing and physically impaired students, such as vibrational feedback, subtitles, and alternative locomotion options. SUS scores are used to determine overall usability, with higher scores indicating a more positive user experience. The questionnaire is shown in Appendix A.

5.2.2. Satisfaction Questionnaire for Assistive Features

The satisfaction questionnaire collects feedback on the accessibility and usability of key features within the VR-ISLS platform. Using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree), it evaluates how well each assistive feature meets the needs of hearing and physically impaired students. Questions focus on the clarity of the user interface layout, the effectiveness of audio cues for navigation, and the readability of on-screen text. Customizable locomotion options, such as movement controls and teleportation, are assessed for their comfort and usability. Additionally, the questionnaire evaluates the role

of the sign language interpreter feature in enhancing comprehension of VR content and instructions. The questionnaire is shown in Appendix B.

5.2.3. Open-Ended Questionnaire

To gather insights into students' perceptions and feedback on the application's effectiveness as a tool for promoting library services tailored to students with impairments, an open-ended questionnaire was administered. We chose an open-ended questionnaire over a semi-structured interview, as this format facilitates more effective communication, especially for participants with hearing impairments who may find interviews challenging. This survey invited participants with varying levels of familiarity with the platform to share their detailed thoughts and experiences with the VR application. By allowing students to freely express their views, the questionnaire aimed to collect valuable information that could guide future improvements and enhance the overall user experience. The complete questionnaire can be found in Appendix C.

5.3. Result and Data Analysis

5.3.1. Result of SUS Scores

To calculate the SUS score, each response is adjusted based on whether the item is positive or negative. For positive (odd-numbered) items, 1 is subtracted from the response score, whereas for negative (even-numbered) items, the response score is subtracted from 5. The adjusted scores are then totaled across all 10 items, and the sum is multiplied by 2.5, resulting in a final SUS score out of 100. In this study, the VR system received an average SUS score of approximately 77 out of 100 from twenty participants, as shown in Figure 7 and Table 7. Based on established research, SUS scores are typically categorized into five levels: A (90–100), B (80–89), C (70–79), D (60–69), and F (0–59) [48,50]. A score of 77 places this VR application in the C range, indicating an acceptable level of usability. According to SUS scoring guidelines, this score falls between “good” and “excellent” [48], suggesting that the system is functional and generally acceptable for impaired students, although there are areas for improvement to enhance user satisfaction. Figure 8 shows the VR system's position within this grading scale.

Table 7. Results of system usability scale.

No.	Type	Questions	N	Mean	SD
1	positive	I think I would like to use the virtual reality system to navigate the environment.	20	4.05	0.70
2	negative	I found the virtual reality system unnecessarily complex.	20	1.94	0.70
3	positive	I thought the virtual reality system was easy to use.	20	3.89	0.65
4	negative	I think I would need support from a technical person to use this virtual reality system.	20	2.10	0.93
5	positive	I found the various assistive functions in the virtual reality system to be well integrated.	20	4.05	0.70
6	negative	I thought there was too much inconsistency in the virtual reality system.	20	2.05	0.84
7	positive	I imagine most impaired students would learn to use the virtual reality system very quickly.	20	4.21	0.71
8	negative	I found the virtual reality system cumbersome to use.	20	2.15	0.76
9	positive	I felt confident using the virtual reality system.	20	4.05	0.70
10	negative	I needed to learn many things before I could start using this system.	20	2.21	0.85

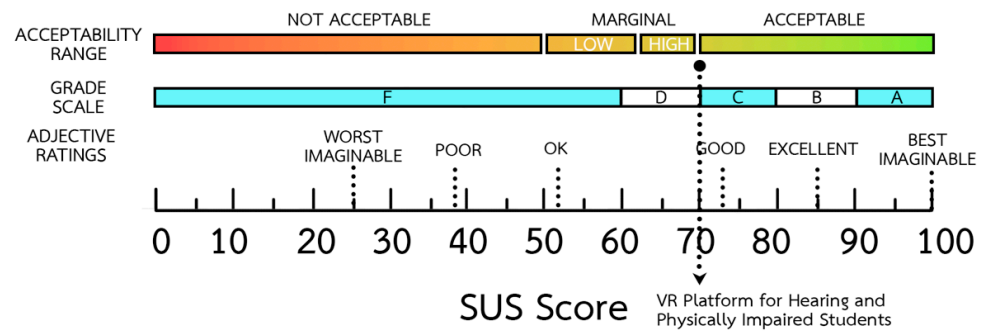


Figure 7. The result of the SUS score for the VR platform.

5.3.2. Result of the Satisfaction Questionnaire

The results of the questionnaire were produced by calculating mean scores and standard deviations for each feature, providing a quantitative measure of user satisfaction. The satisfaction questionnaire results, shown in Figure 8 and Table 8, indicate generally positive ratings for the VR platform's assistive features. The user interface received a mean score of 3.35 with an SD of 0.98, while audio cues were rated slightly higher, with a mean of 3.45 and an SD of 1.14. It is important to note that two participants did not rate the audio cues feature because they were completely deaf and unable to use this functionality. The text display feature achieved a mean score of 3.80 (SD = 1.05), reflecting favorable perceptions. Vibration feedback scored highly, with a mean of 4.15 and an SD of 0.98, indicating strong approval. Locomotion control was also well-received, with a mean score of 3.95 and an SD of 0.88. The sign language interpreter, assessed by a smaller group of participants, received the highest satisfaction rating, with a mean of 4.25 and an SD of 0.50. This feature was evaluated by only four hearing-impaired participants who understood sign language. Based on the overall results, vibration feedback and the sign language interpreter were the features with the highest satisfaction ratings.

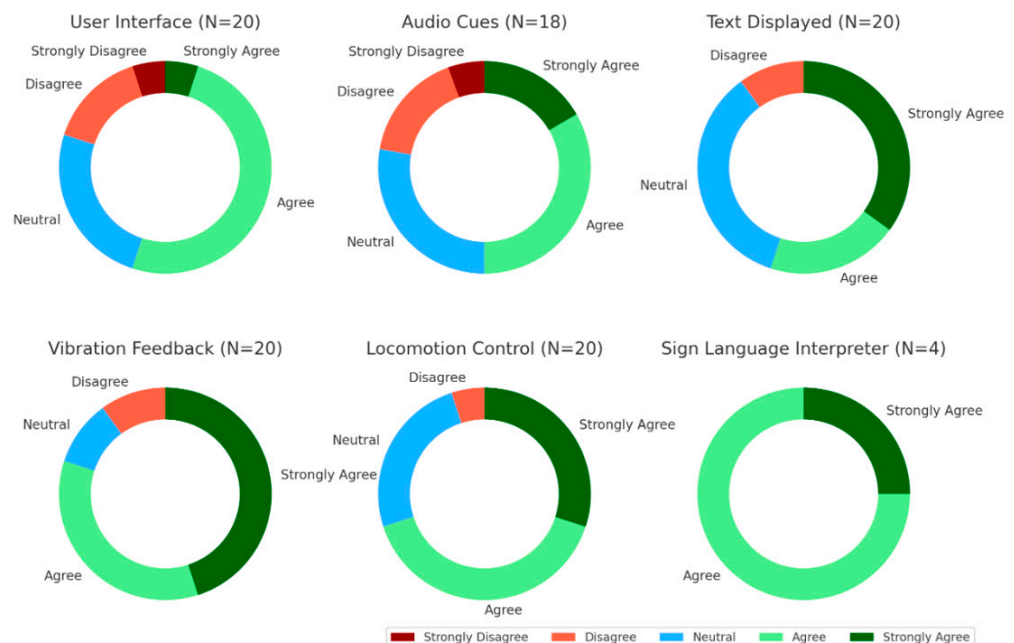


Figure 8. The result of satisfaction questionnaire.

Table 8. Results of satisfaction questionnaire for assistive features.

Question No.	Assistive Features	N	Mean	SD
1	user interface	20	3.35	0.98
2	audio cues	18 *	3.45	1.14
3	text displayed	20	3.80	1.05
4	Vibration feedback	20	4.15	0.98
5	Locomotion control	20	3.95	0.88
6	sign language interpreter	4 **	4.25	0.50

* Excludes participants with complete hearing loss. ** Excludes participants who do not understand sign language.

5.3.3. Result of Open-Ended Questionnaire

- Feedback on Visual Environment

Participants appreciated the virtual library's realistic representation, which allowed them to explore the layout and become familiar with it, helping to alleviate spatial-related anxiety and build confidence. Many comments highlighted the realistic feel of the VR environment, which could be beneficial for impaired students who experience anxiety in the physical library. One participant noted, "The VR environment made it easy to recognize where things are, which makes me feel more confident when I'm actually in the library". The immersive environment enabled easy identification of key sections and resources, while the intuitive VR controls facilitated smooth navigation without reports of motion sickness.

While participants found these aspects beneficial, they suggested further enhancements to improve practicality and engagement. Some recommended adding AI-driven NPCs (non-player characters) to interact with users, answer questions, and offer guidance. One participant remarked, "It would be great if there were virtual librarians who could answer questions or point me in the right direction". Others proposed small educational games within the VR library to make learning more enjoyable, covering topics relevant to library skills. Another participant commented, "Adding mini-games or interactive challenges would make the experience more fun and help me learn about library resources in a different way". Participants also expressed interest in features such as borrowing books, accessing digital materials, and receiving personalized book recommendations. As one user stated, "If I could borrow e-books or get recommendations based on what I'm interested in, it would make the VR library even more useful".

- Feedback on Assistive Functions

The VR system's assistive functions received mixed feedback, with particular appreciation for the user interface and text display options, though participants highlighted areas for improvement. While the user interface was generally clear and intuitive, facilitating navigation for users with different levels of familiarity, some participants desired additional customization options. For example, they wanted the ability to display only Thai text without English to reduce visual clutter. Others noted that the layout proportions were suboptimal, suggesting that "visual materials should be larger and more prominent, while the text could be smaller". The text display feature, though helpful for users with visual limitations, could also benefit from more customization; participants expressed interest in adjusting font style, making text bold, or changing colors to suit their preferences. One participant remarked, "The adjustable text sizes really helped me read comfortably, but it would be even better if I could further customize it". These refinements would help make the VR system more accessible and tailored to individual needs, creating a more inclusive experience overall.

In terms of audio cues, hearing-impaired participants, particularly those using hearing aids, found the existing auditory features valuable but recommended further customization to improve usability. They emphasized the need for a noise reduction feature, adjustable

sound levels to accommodate users with varying degrees of residual hearing, and the option to adjust the speed of audio cues, allowing playback at 2× or 3× speeds for quicker comprehension. One participant noted, “Having the option to adjust sound levels would help me use the system more effectively”. Additionally, participants suggested incorporating auditory guides or directional cues to assist both hearing-impaired and visually impaired users with navigation. As one user explained, “Sound cues would be a great help in orienting myself within the VR library”. Furthermore, vibration feedback was highlighted as a beneficial feature, providing tactile cues for interaction points and assisting users who may miss auditory signals. These combined audio and vibration adjustments, along with the flexibility to modify audio cue speed, would create a more supportive and accessible environment for those who rely on different sensory feedback to navigate.

One of the most appreciated assistive functions was the sign language interpreter integrated into the VR environment, which facilitated smoother navigation for hearing-impaired participants. However, some users noted that the interpreter’s speed could be challenging, with one participant commenting, “If I could adjust the interpreter speed, it would be easier to follow along”. The locomotion control features, such as gamepad-based movement and teleportation, were also highly valued, especially by mobility-impaired users. Participants suggested options for single-handed controls and minimized head movement requirements. As one user noted, “The teleportation feature was great, but single-handed control would make it even easier”. Collectively, these recommendations underscore the potential for further refinement, highlighting the value of customizable assistive features to create a more inclusive and engaging VR experience.

6. Discussion and Finding

6.1. Designed for Usability in the VR Environment

The VR-ISLS platform was specifically designed to explore how immersive virtual reality can effectively support hearing and physically impaired students in navigating the main university library (RQ1). Research indicates that these students often experience heightened anxiety in traditional library settings due to accessibility and mobility challenges [51]. To address these issues, the VR-ISLS development process focused on creating a realistic virtual environment that replicates the library’s layout, resources, and services, incorporating input from disability support specialists, librarians, VR developers, and health experts. This collaboration was essential for tailoring the platform’s assistive features, such as sign language interpretation, customizable audio cues, vibration feedback near interaction points, and multiple options for accessible navigation controls with minimized head movement. These features aimed to help students become familiar with the library space before visiting, reducing anxiety by providing a safe, controlled environment for practicing navigation skills. Based on our implementation experience, collaboration with health experts and disability support specialists provided crucial insights into the unique needs of impaired students, who often have specific concerns and sensitivities. Including these professionals in the design team is recommended for developing VR environments that are truly supportive and inclusive. The design aligns with related studies [52,53], which show that VR applications can promote confidence and reduce stress by offering a supportive, low-pressure space for building navigation skills and increasing familiarity with the environment.

To assess the usability of the VR environment (RQ2), the VR system’s design aimed to prioritize usability by creating an intuitive, accessible, and visually supportive space for all users, particularly those with impairments. User feedback indicated that participants generally found the VR interface clear and user-friendly, enabling effective navigation and interaction with the virtual library, which helped reduce anxiety and increase familiarity.

This positive response was reflected in the System Usability Scale (SUS) score, averaging approximately 77, indicating good usability and alignment with general user expectations. Qualitative feedback from impaired participants highlighted the need for additional customization options [54] to enhance accessibility and comfort. Research by [55] supports that features such as haptic feedback and customizable visuals enhance usability for visually impaired users, while [56] found that contrast and text size adjustments are highly beneficial, aligning with participant preferences for improved readability and reduced cognitive load. Further studies by [57,58] confirm that adaptive interfaces and customization features significantly improve VR usability and inclusivity, making the environment more approachable. Additionally, [59] recommends customizable assistive options for hearing-impaired users to facilitate communication and reduce anxiety, reinforcing the theme of inclusivity. Collectively, these studies validate the effectiveness of adaptable VR systems in fostering accessibility, reducing anxiety, and creating a more personalized, welcoming environment.

6.2. Assistive Functions of VR

The essential design of assistive functions for hearing and physically impaired students (RQ3) was well-received, providing valuable support for participants with hearing, visual, and mobility impairments. Feedback on specific features, such as audio cues, vibration feedback, and sign language interpretation, highlighted the potential of VR as an accessible platform when assistive technology is thoughtfully integrated. For hearing-impaired users, audio cues were beneficial, though further customization was suggested, including noise reduction, adjustable volume, and playback speeds of $2\times$ or $3\times$ for faster comprehension. Vibration feedback emerged as a crucial feature, providing tactile cues for interaction points and proving especially useful for users who may miss auditory signals. The sign language interpreter was also praised for aiding navigation, though participants recommended customizable interpreter speeds to suit different levels of sign language proficiency. Locomotion control functions, including gamepad-based movement and teleportation, were valued by mobility-impaired users, who suggested single-handed control options and minimized head movement to further enhance accessibility.

Based on our experience during implementation and qualitative feedback, we found that impaired students have varied requirements and frequently request different customization options. Thus, it is challenging to recommend a universal assistive feature for all hearing and physically impaired students. However, we can provide recommendations for those designing VR environments for these users. For students with hearing impairments, it is crucial to develop a system that includes sign language interpretation videos for each information point [60], especially for those with profound hearing loss. Allowing users to adjust the speed of these interpretation videos would significantly enhance accessibility and overall user experience. Feedback from students with partial hearing loss highlighted a significant concern: the inconvenience of having to remove their hearing aids each time they use the system. They also expressed a preference for functionalities that enable customization of subtitle clarity [61], sound adjustments, and background noise reduction [62].

For students with physical impairments, the most requested assistive feature was an alternative locomotion controller [63]. Observations and feedback indicated that most students, including those with muscle weakness from muscular dystrophy or those using wheelchairs, did not encounter significant issues when using VR. A notable exception was a student with an upper limb amputation who could only use one hand. She expressed the need for specialized controls that allow one-handed customization, presenting a unique challenge. Unfortunately, there is a lack of existing research on designing VR controls specifically for one-handed use. Additionally, three students in this group reported experi-

encing cybersickness, which might be associated with muscular dystrophy. This aligns with findings from [64,65], suggesting that comfort levels related to sensory mismatch during VR use while in a wheelchair could be contributing factors.

7. Conclusions

This study focused on designing and developing the VR-ISLS platform to support hearing and physically impaired students in navigating and utilizing library services at Chiang Mai University. The findings demonstrated that thoughtfully integrated assistive functions, such as sign language interpretation, audio cues, vibration feedback, and customizable locomotion controls, can significantly enhance accessibility and usability. Feedback from participants confirmed the platform's potential to reduce anxiety and promote confidence by familiarizing students with the library environment in a controlled, virtual setting. The positive System Usability Scale (SUS) score of approximately 77 reinforced the platform's effective design, suggesting that it meets user expectations for usability.

However, the study also highlighted areas for future improvement, such as the inclusion of more flexible customization options for audio and visual settings, tailored sign language interpreter speeds, and alternative control schemes for users with specific physical limitations. Addressing these enhancements will ensure that the VR platform can cater to a wider range of individual needs, promoting inclusivity and providing a supportive tool that aligns with accessibility standards. Moving forward, continuous user feedback and iterative development will be key to refining the VR-ISLS platform and extending its capabilities as an inclusive educational tool for all impaired students.

Author Contributions: Conceptualization, P.A. and K.P.; methodology, P.A. and K.P.; software, Y.Y.; validation, Y.Y. and K.I.; formal analysis, P.A. and K.I.; investigation, Y.Y.; resources, Y.Y.; data curation, K.I.; writing—original draft preparation, P.T. and K.P.; writing—review and editing, K.P.; visualization, Y.Y.; supervision, K.P.; project administration, Y.Y.; funding acquisition, K.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research project was supported by the Fundamental Fund 2025, Chiang Mai University, and also Thailand Science Research and Innovation (TSRI) 2025.

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of the Chiang Mai University Research Ethics Committee (CMU REC No. 66/169).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study are available upon request from the corresponding author due to restrictions. The data are not publicly available.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

1. I think I would like to use the virtual reality system to navigate the environment.
2. I found the virtual reality system unnecessarily complex.
3. I thought the virtual reality system was easy to use.
4. I think I would need support from a technical person to use this virtual reality system.
5. I found the various assistive functions in the virtual reality system to be well integrated.
6. I thought there was too much inconsistency in the virtual reality system.
7. I imagine most impaired students would learn to use the virtual reality system very quickly.
8. I found the virtual reality system cumbersome to use.
9. I felt confident using the virtual reality system.

10. I needed to learn many things before I could start using this system.

Appendix B

1. The user interface was clearly laid out, making it easy for me to navigate and locate features.
2. The audio cues in the VR environment were helpful and clear, enhancing my ability to learn about the virtual library.
3. The text displayed in the VR environment was easy to read and understand, allowing me to access information without difficulty.
4. The customization options for locomotion (e.g., movement control and teleportation) allowed me to choose a comfortable way to navigate the VR environment.
5. The inclusion of a sign language interpreter feature effectively supported my understanding of the VR content and instructions.

Appendix C

1. What was your experience with the usability and accessibility of the VR platform?
2. What did you find most beneficial about the assistive features in the VR platform, and how could they be improved to better meet your needs?
3. Do you have any additional suggestions for enhancing the VR platform?

References

1. Rupp, M.A.; Odette, K.L.; Kozachuk, J.; Michaelis, J.R.; Smither, J.A.; Daniel, S.; McConnell, D.S. Investigating learning outcomes and subjective experiences in 360-degree videos. *Comput. Educ.* **2019**, *128*, 256–268. [\[CrossRef\]](#)
2. Haj-Bolouri, A.; Rossi, M. Towards design principles for safety training in virtual reality: An action design research case. In *International Conference on Design Science Research in Information Systems and Technology*; Springer: Cham, Switzerland, 2021; pp. 89–95.
3. Schutte, N.S.; Stilinović, E.J. Facilitating empathy through virtual reality. *Motiv. Emot.* **2017**, *41*, 708–712. [\[CrossRef\]](#)
4. Herrera, F.; Bailenson, J.; Weisz, E.; Ogle, E.; Zaki, J. Building long-term empathy: A large-scale comparison of traditional and virtual reality perspective-taking. *PLoS ONE* **2018**, *13*, e0204494. [\[CrossRef\]](#) [\[PubMed\]](#)
5. Didehbani, N.; Allen, T.; Kandalaft, M.; Krawczyk, D.; Chapman, S. Virtual reality social cognition training for children with high-functioning autism. *Comput. Hum. Behav.* **2016**, *62*, 703–711. [\[CrossRef\]](#)
6. Renault, H. Virtual Reality Is Helping People with Disabilities Conquer Overwhelming Real-Life Situations. ABC News, 7 July 2018. Available online: <https://www.abc.net.au/news/2018-07-07/virtual-reality-people-with-disabilities-practise-life-skills/9947740> (accessed on 18 June 2024).
7. Lotan, M. Training caregivers to provide virtual reality intervention for adults with severe intellectual and developmental disability. *J. Phys. Ther. Educ.* **2011**, *25*, 15–19. [\[CrossRef\]](#)
8. Jiang, Z.; Meltzer, A.; Zhang, X. Using virtual reality to implement disability studies' advocacy principles: Uncovering the perspectives of people with disability. *Disabil. Soc.* **2023**, *39*, 1592–1612. [\[CrossRef\]](#)
9. Varnum, K.J. (Ed.) *Beyond Reality: Augmented, Virtual, and Mixed Reality in the Library*; ALA Editions: Chicago, IL, USA, 2019.
10. LaValle, S.M. The physiology of human vision. In *Virtual Reality*; Cambridge University Press: Cambridge, UK, 2019; pp. 128–152.
11. Kim, S.H.; Choi, G.W.; Jung, Y.J. Design principles for transforming making programs into online settings at public libraries. *Inf. Learn. Sci.* **2020**, *121*, 619–630. [\[CrossRef\]](#)
12. Subramaniam, M.; Scaff, L.; Kawas, S.; Hoffman, K.M.; Davis, K. Using technology to support equity and inclusion in youth library programming: Current practices and future opportunities. *Libr. Q.* **2018**, *88*, 315–331. [\[CrossRef\]](#)
13. Huang, C.L.; Luo, Y.F.; Yang, S.C.; Lu, C.M.; Chen, A.S. Influence of students' learning style, sense of presence, and cognitive load on learning outcomes in an immersive virtual reality learning environment. *J. Educ. Comput. Res.* **2020**, *58*, 596–615. [\[CrossRef\]](#)
14. Oliver, M. The social model of disability: Thirty years on. *Disabil. Soc.* **2013**, *28*, 1024–1026. [\[CrossRef\]](#)
15. Krishnan, I.A.; De Mello, G.; Kok, S.A.; Sabapathy, S.K.; Munian, S.; Ching, H.S.; Kanan, V.N. Challenges faced by hearing impairment students during COVID-19. *Malays. J. Soc. Sci. Humanit. (MJSSH)* **2020**, *5*, 106–116. [\[CrossRef\]](#)
16. World Health Organization. Deafness and Hearing Loss. 2024. Available online: <https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss> (accessed on 18 June 2024).

17. James, M.; P, K.; Ninan, P.J. A study on prevalence and risk factors of hearing impairment among newborns. *Int. J. Contemp. Pediatr.* **2018**, *5*, 304. [CrossRef]
18. Moctar, E.C.M.; Riahi, Z.; El Hachmi, H.; Vetten, F.; Meiloud, G.; Bonnet, C.; Abdelhak, S.; Errami, M.; Houmeida, A. Etiology and associated GJB2 mutations in Mauritanian children with non-syndromic hearing loss. *Eur. Arch. Otorhinolaryngol.* **2016**, *273*, 3693–3698. [CrossRef] [PubMed]
19. Chung, E.; Subramaniam, G.; Christ, D.L. Online learning readiness among university students in Malaysia amidst COVID-19. *Asian J. Univ. Educ.* **2020**, *16*, 47–58. [CrossRef]
20. Verbrugge, L.M.; Jette, A.M. The disablement process. *Soc. Sci. Med.* **1994**, *38*, 1–14. [CrossRef] [PubMed]
21. Konczal, E.F. (Ed.) *Physical Disabilities*; Rutgers University, Department of Kinesiology and Health: Rutgers, NJ, USA, 2024. Available online: <https://kines.rutgers.edu/dshw/disabilities/physical/1060-physical-disabilities> (accessed on 18 June 2024).
22. Berg, V. Types of Physical Disabilities. 2024. Available online: <https://www.carehome.co.uk/advice/types-of-physical-disabilities> (accessed on 18 June 2024).
23. Department of Health & Human Services. *Physical Disabilities*; Better Health Channel: Victoria, Australia, 2015. Available online: <https://www.betterhealth.vic.gov.au/health/servicesandsupport/physical-disabilities> (accessed on 18 June 2024).
24. Lebrasseur, A.; Fortin-Bédard, N.; Lettre, J.; Bussi eres, E.L.; Best, K.; Boucher, N.; Hotton, M.; Beaulieu-Bonneau, S.; Mercier, C.; Lamontagne, M.E.; et al. Impact of COVID-19 on people with physical disabilities: A rapid review. *Disabil. Health J.* **2021**, *14*, 101014. [CrossRef]
25. Autry, C.E.; Anderson, S.C. Virtual reality sailing simulation for people with physical disabilities: Learn to sail. *Ther. Recreat. J.* **2022**, *56*, 177–190. [CrossRef]
26. Kepp, N.E.; Arrieta, I.; Schi oth, C.; Percy-Smith, L. Virtual Reality pitch ranking in children with cochlear implants, hearing aids or normal hearing. *Int. J. Pediatr. Otorhinolaryngol.* **2022**, *161*, 111241. [CrossRef] [PubMed]
27. Lai, B.; Young, R.; Craig, M.; Chaviano, K.; Swanson-Kimani, E.; Wozow, C.; Davis, D.; Rimmer, J.H. Improving social isolation and loneliness among adolescents with physical disabilities through Group-Based Virtual Reality Gaming: Feasibility Pre-Post Trial study. *JMIR Form. Res.* **2023**, *7*, e47630. [CrossRef] [PubMed]
28. Intawong, K.; Olson, D.; Chariyalertsak, S. Application technology to fight the COVID-19 pandemic: Lessons learned in Thailand. *Biochem. Biophys. Res. Commun.* **2021**, *538*, 231–237. [CrossRef] [PubMed]
29. Kannegiser, S. Effects of an augmented reality library orientation on anxiety and self-efficacy: An exploratory study. *Coll. Res. Libr.* **2021**, *82*, 352. [CrossRef]
30. Sureephong, P.; Chernbumroong, S.; Intawong, K.; Jansukpum, K.; Wongwan, N.; Puritat, K. The effect of virtual reality on knowledge acquisition and situational interest regarding library orientation in the time of COVID-19. *J. Acad. Librariansh.* **2023**, *49*, 102789. [CrossRef]
31. Zhao, J.C.Y.; Mawhinney, T. Effectiveness of vernacular library orientation videos in comparison with the English language equivalent. *Coll. Res. Libr.* **2020**, *81*, 1107. [CrossRef]
32. Hickner, A.; Wright, D.; Merlo, L.; Gordon-Elliott, J.S.; Delgado, D. Redesigning library orientation for first-year medical students during the pandemic. *J. Med. Libr. Assoc. JMLA* **2021**, *109*, 497–502. [CrossRef] [PubMed]
33. Dozio, N.; Marcolin, F.; Scurati, G.W.; Ulrich, L.; Nonis, F.; Vezzetti, E.; Marsocci, G.; La Rosa, A.; Ferrise, F. A design methodology for affective virtual reality. *Int. J. Hum.-Comput. Stud.* **2022**, *162*, 102791. [CrossRef]
34. Opoku, D.G.J.; Perera, S.; Osei-Kyei, R.; Rashidi, M.; Bamdad, K.; Famakinwa, T. Digital twin for indoor condition monitoring in living labs: University library case study. *Autom. Constr.* **2024**, *157*, 105188. [CrossRef]
35. Liu, M.; Fang, S.; Dong, H.; Xu, C. Review of digital twin about concepts, technologies, and industrial applications. *J. Manuf. Syst.* **2021**, *58*, 346–361. [CrossRef]
36. Melo, R.S.; Lemos, A.; Delgado, A.; Raposo, M.C.F.; Ferraz, K.M.; Belian, R.B. Use of virtual reality-based games to improve balance and gait of children and adolescents with sensorineural hearing loss: A systematic review and meta-analysis. *Sensors* **2023**, *23*, 6601. [CrossRef]
37. Kwak, C.; Han, W.; Bahng, J. Systematic review and meta-analysis of the application of virtual reality in hearing disorders. *J. Audiol. Otol.* **2022**, *26*, 169–181. [CrossRef] [PubMed]
38. Aliakbaryhosseinabadi, S.; Keidser, G.; May, T.; Dau, T.; Wendt, D.; Rotger-Griful, S. The effects of noise and simulated conductive hearing loss on physiological response measures during interactive conversations. *J. Speech Lang. Hear. Res.* **2023**, *66*, 4009–4024. [CrossRef] [PubMed]
39. Maier, H.; Lenarz, T.; Agha-Mir-Salim, P.; Agterberg, M.J.; Anagiotos, A.; Arndt, S.; Ball, G.; Bance, M.; Barbara, M.; Baumann, U.; et al. Consensus statement on bone conduction devices and active middle ear implants in conductive and mixed hearing loss. *Otol. Neurotol.* **2022**, *43*, 513–529.
40. Hermawati, S.; Pieri, K. Assistive technologies for severe and profound hearing loss: Beyond hearing aids and implants. *Assist. Technol.* **2020**, *32*, 182–193. [CrossRef] [PubMed]

41. Palaniappan, S.M.; Zhang, T.; Duerstock, B.S. Identifying comfort areas in 3D space for persons with upper extremity mobility impairments using virtual reality. In Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility, Pittsburgh, PA, USA, 28–30 October 2019; pp. 495–499.
42. Ülger, Ö.; Yıldırım Şahan, T.; Celik, S.E. A systematic literature review of physiotherapy and rehabilitation approaches to lower-limb amputation. *Physiother. Theory Pract.* **2018**, *34*, 821–834. [\[CrossRef\]](#)
43. Kurt-Aydin, M.; Savaş-Kalender, D.; Tarsuslu, T.; Yis, U. Feasibility of virtual reality and comparison of its effectiveness to biofeedback in children with Duchenne and Becker muscular dystrophies. *Muscle Nerve* **2024**, *70*, 82–93. [\[CrossRef\]](#) [\[PubMed\]](#)
44. Pepe, M.; Costantino, D. Techniques, tools, platforms and algorithms in close range photogrammetry in building 3D model and 2D representation of objects and complex architectures. *Comput.-Aided Des. Appl.* **2020**, *18*, 42–65. [\[CrossRef\]](#)
45. Kingsland, K. Comparative analysis of digital photogrammetry software for cultural heritage. *Digit. Appl. Archaeol. Cult. Herit.* **2020**, *18*, e00157. [\[CrossRef\]](#)
46. Zhao, J.; An, R.; Xu, R.; Lin, B. Comparing hand gestures and a gamepad interface for locomotion in virtual environments. *Int. J. Hum.-Comput. Stud.* **2022**, *166*, 102868. [\[CrossRef\]](#)
47. Nasiri, M.; Porter, J.; Kohm, K.; Robb, A. Changes in navigation over time: A comparison of teleportation and joystick-based locomotion. *ACM Trans. Appl. Percept.* **2023**, *20*, 16. [\[CrossRef\]](#)
48. Brooke, J. SUS: A quick and dirty usability scale. In *Usability Evaluation in Industry*; Jordan, P.W., Thomas, B., Weerdmeester, B.A., McClelland, I.L., Eds.; Taylor & Francis: Abingdon, UK, 1995; pp. 189–194.
49. Huang, H.; Lee, C.F. Factors affecting usability of 3D model learning in a virtual reality environment. *Interact. Learn. Environ.* **2022**, *30*, 848–861. [\[CrossRef\]](#)
50. Bangor, A.; Kortum, P.; Miller, J. Determining what individual SUS scores mean: Adding an adjective rating scale. *J. Usability Stud.* **2009**, *4*, 114–123.
51. Ahmed, M.R.; Naveed, M.A. Information accessibility for visually impaired students. *Pak. J. Inf. Manag. Libr.* **2021**, *22*, 16–36. [\[CrossRef\]](#) [\[PubMed\]](#)
52. Taçgın, Z. The perceived effectiveness regarding immersive virtual reality learning environments changes by the prior knowledge of learners. *Educ. Inf. Technol.* **2020**, *25*, 2791–2809. [\[CrossRef\]](#)
53. Buttussi, F.; Chittaro, L. A comparison of procedural safety training in three conditions: Virtual reality headset, smartphone, and printed materials. *IEEE Trans. Learn. Technol.* **2020**, *14*, 1–15. [\[CrossRef\]](#)
54. Ríos-Hernández, M.; Jacinto-Villegas, J.M.; Portillo-Rodríguez, O.; Vilchis-González, A.H. User-centered design and evaluation of an upper limb rehabilitation system with a virtual environment. *Appl. Sci.* **2021**, *11*, 9500. [\[CrossRef\]](#)
55. Keskinen, T. Evaluating the User Experience of Interactive Systems in Challenging Circumstances. Doctoral Dissertation, University of Tampere, Tampere, Finland, 2015. Available online: <https://trepo.tuni.fi/bitstream/handle/10024/98132/978-951-44-9972-2.pdf;jsessionid=730243B9DF8A3229A352CC9F8CAA43DF?sequence=1> (accessed on 18 June 2024).
56. Theodorou, P.; Tsiligkos, K.; Meliones, A.; Filios, C. An extended usability and UX evaluation of a mobile application for the navigation of individuals with blindness and visual impairments outdoors—An evaluation framework based on training. *Sensors* **2022**, *22*, 4538. [\[CrossRef\]](#) [\[PubMed\]](#)
57. Zouhaier, L.; BenDalyHlaoui, Y.; Ayed, L.B. Adaptive user interface based on accessibility context. *Multimed. Tools Appl.* **2023**, *82*, 35621–35650. [\[CrossRef\]](#)
58. Tuena, C.; Pedroli, E.; Trimarchi, P.D.; Gallucci, A.; Chiappini, M.; Goulene, K.; Gaggioli, A.; Riva, G.; Lattanzio, F.; Giunco, F.; et al. Usability issues of clinical and research applications of virtual reality in older people: A systematic review. *Front. Hum. Neurosci.* **2020**, *14*, 93. [\[CrossRef\]](#) [\[PubMed\]](#)
59. Kim, H.; Hwang, H.; Gwak, S.; Yoon, J.; Park, K. Improving communication and promoting social inclusion for hearing-impaired users: Usability evaluation and design recommendations for assistive mobile applications. *PLoS ONE* **2024**, *19*, e0305726. [\[CrossRef\]](#)
60. Anderton, C. Investigating sign language interpreter rendering and guiding methods in virtual reality 360-degree content. In Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility, Athens Greece, 23–26 October 2022; pp. 1–6.
61. Teófilo, M.; Lourenço, A.; Postal, J.; Lucena, V.F. Exploring virtual reality to enable deaf or hard of hearing accessibility in live theaters: A case study. In *Universal Access in Human-Computer Interaction. Virtual, Augmented, and Intelligent Environments*; Lecture Notes in Computer Science; Springer: Cham, Switzerland, 2018; pp. 132–148. [\[CrossRef\]](#)
62. Serafin, S.; Adjorlu, A.; Percy-Smith, L.M. A review of virtual reality for individuals with hearing impairments. *Multimodal Technol. Interact.* **2023**, *7*, 36. [\[CrossRef\]](#)
63. Franz, R.L.; Yu, J.; Wobbrock, J.O. Comparing locomotion techniques in virtual reality for people with upper-body motor impairments. In Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility, New York, NY, USA, 22–25 October 2023; ACM: New York, NY, USA, 2023; pp. 1–15.

64. Aykent, B.; Paillot, D.; Mérienne, F.D.R.; Fang, Z.; Kemeny, A. Study of the influence of different washout algorithms on simulator sickness for a driving simulation task. In *ASME 2011 World Conference on Innovative Virtual Reality*; ASME: New York, NY, USA, 2011; Volume 44328, pp. 331–341.
65. Taheri, A.; Caetano, A.; Sra, M. Virtual steps: The experience of walking for a lifelong wheelchair user in virtual reality. In *Proceedings of the 2024 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, Orlando, FL, USA, 16–21 March 2024; IEEE: Piscataway, NJ, USA, 2024; pp. 168–178.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.